LECTURE 39: MATH1061/7861
TOPICS: FINAL EXAM REVISION
DATE: THURSDAY MAY 28, 2026

Two help sessions for the final:

~~One Thursday June~~

- Friday June 5, 11:00 am – 1:00 p.m 50-N201

- Thursday June 11, 11:00 am – 1:00 p.m 50-N201

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13 Questions on the Final.

① Logic: I give you a statement involving $p, q, r, \ldots$ And ask you whether it is true or not (whether the argument is valid).

↳ Always justify your answer.

2. Quantified & conditional statements and their negation.
   - Decide whether the original statement or negation is true. Justify!

3. GCD & prime factorisation.
   - Euclidean Algorithm.

4. (Recursive) sequences and (Strong) induction.

5. Set theory (intersection, union, product, Power set).

6. Functions (Domain, Codomain, image, fibre, surjective, injective, ...)

(7) Relations → Equivalence Relation.
→ Partial orders

(8) Cardinality
↳ ~~Do~~ Give you two sets, and ask whether they have the same cardinality.
→ No uncountable business.

(9) Groups & subgroups.
↳ Cayley table.
↳ $(\mathbb{Z}_n, +)$, $(\mathbb{Z}_p - \{0\}, \times)$
→ $(\mathbb{Q}, +)$, $(\mathbb{R}, +)$, ...

→ No associativity

(10) Fields $(F, +, \times, 0, 1)$

Example: $\mathbb{R}, \mathbb{Q}, \mathbb{Z}_p$

↳ solve linear equation.

(11) Binomial coefficients $\binom{n}{k}$ $(x+y)^n$.

(12) Counting & probability.

↳ multi-part.

↳ inclusion / exclusion.

(13) Graph theory

↳ Eulerian circuit, trace, Adjancy matrix, Handshake theorem, ...